Erode Mandi Price Volatility & Market Opportunity Dashboard (2021–2025)

1. Objective:

This mini project focuses on agricultural market data analysis using Microsoft Excel and Power BI on Erode district mandi price data from 2021 to 2025. It includes cleaning and transforming raw market data, developing a dimensional data model, and building an interactive dashboard to analyze commodity price trends, volatility, market-level pricing patterns, and potential selling opportunities.

2. Data Source:

- **Source: Open Government Data Platform India / AGMARKNET mandi price data**
- **Dataset: Variety-wise daily market prices data for agricultural commodities**
- **Timeline: 2021–2025**
- **Geographic Scope: Erode district, Tamil Nadu**
- **Domain: Agriculture / Market Intelligence**

The dataset contains daily mandi-level agricultural price observations including commodity name, variety, grade, market, district, date, and minimum, maximum, and modal prices. This makes it suitable for time-series analysis, commodity-level comparison, and market opportunity analysis.

3. Business Problem Statement:

Agricultural commodity prices vary significantly across markets, varieties, and time periods. Farmers, traders, and market analysts often lack a simple visual system to identify stable markets, volatile commodities, seasonal opportunities, and favourable selling locations. This project addresses that gap by analyzing Erode mandi price data to identify price spread patterns, commodity volatility, premium markets, and time-based opportunity signals.

The project specifically aims to:

- Analyze commodity-wise price volatility
- Identify high-price and low-price markets
- Study seasonal and monthly price movement
- Compare modal price and price spread across markets
- Identify stable vs unstable markets
- Build an interactive dashboard for decision-oriented market intelligence

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Arrival_Date	Date	Date on which the mandi price record was captured. Used to connect the fact table with the Date table.
Commodity	Text	Name of the commodity sold in the market, such as Garlic, Jasmine, or Dry Chillies.
Commodity_Code	Whole Number / Text	Unique code assigned to the commodity. Used to link the fact table with the commodity dimension.
District	Text	District in which the mandi market is located. In this project, mainly Erode district data is analyzed.
Grade	Text	Quality classification of the commodity, such as FAQ or local grade.
Market	Text	Name of the mandi / market where the commodity was traded.
Max_Price	Decimal Number / Whole Number	Highest price recorded for the commodity in that market on that date.

Min_Price	Decimal Number / Whole Number	Lowest price recorded for the commodity in that market on that date.
Modal_Price	Decimal Number / Whole Number	Most common or prevailing price recorded for the commodity in that market on that date. This is the primary price used in analysis.
State	Text	State in which the mandi market is located.
Variety	Text	Specific variety/type of the commodity, such as local or specific cultivar.

Among these attributes, Modal_Price is the primary measure used for trend and opportunity analysis, while Min_Price and Max_Price are used to derive price spread and volatility indicators.

5. Tools & Technologies:

5.1 Microsoft Excel:

- Data Cleaning
- Data Transformation
- Standardizing text values
- Creating month and year columns
- Removing duplicates

5.2 Power BI:

- Data modelling
- DAX
    - Calculated Columns
    - Calculated Measures
    - Time DAX operations
- Interactive Dashboard Creation and Visualization
- Slicer-based filtering
- Time trend analysis
- Market and commodity comparison

6. Data Pre-Processing (Excel / Power Query):

6.1 Data Cleaning & Transformation:

1) Converted Arrival_Date into proper date format.
2) Converted price columns into numeric format.
3) Applied Clean and Trim functions to text fields such as Commodity, Market, Variety, and Grade.
4) Standardized market naming variations wherever necessary.
5) Checked and removed duplicate records.
6) Verified missing values and handled inconsistent text entries.

6.2 Derived Columns:

Additional calculated columns such as Price Spread, Price Band, and Spread Category were created to strengthen the dashboard analysis. These calculated columns were also added as slicers across the dashboard pages to support interactive filtering based on price level and spread intensity.

1) Extracted **Year** from Arrival_Date.
2) Extracted **Month Name** and **Month Number**.
3) Created **Quarter** for quarterly trend analysis.
4) Created **Price Spread** to capture row-level variation between max and min mandi prices
5) Created **Price Band** to classify records into broad price groups.

6) Created **Spread Category** to classify records based on spread intensity for easier volatility interpretation.

6.3 Data Validation:

Basic validation checks were carried out to improve data reliability:

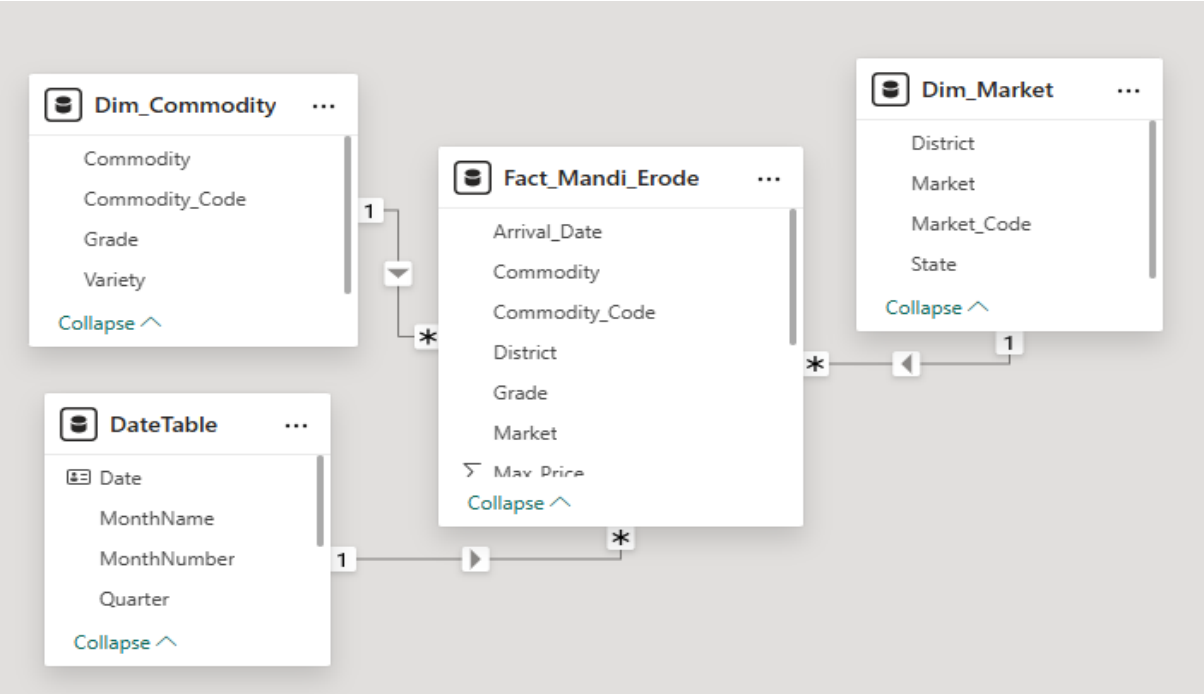
1) Verified that Min_Price <= Modal_Price <= Max_Price wherever applicable.
2) Reviewed unusual values and outliers.
3) Checked market-level naming consistency.

A validation check was performed in Power Query to ensure that all records followed the logical sequence Min_Price <= Modal_Price <= Max_Price. During this process, 13 invalid records were identified and excluded from the final analytical dataset to maintain data consistency and improve the reliability of spread-based analysis.

6.4 Data Modelling:

The Power BI model was designed using a star-schema-style structure with one fact table and three supporting dimension tables. This modelling approach improves filtering, aggregation, and dashboard performance while supporting commodity-wise, market-wise, and time-based analysis.

Figure 1: Data Model



The dataset was structured using a star-schema-style model.

Fact Table: Fact_Mandi_Erode

Dimension Tables: Dim_Commodity, Dim_Market, DateTable

6.4.1 Fact Table:

The fact table contains transactional market price observations such as Arrival_Date, Commodity_Code, Market, Min_Price, Max_Price, and Modal_Price. Additional calculated columns such as Price Spread, Price Band, and Spread Category were created to support volatility interpretation and dashboard filtering.

6.4.2 Dimension Tables:

Dim_Commodity : Stores descriptive information such as Commodity, Commodity_Code, Grade, and Variety.

Dim_Market :Stores State, District, Market, and Market_Code.

DateTable: Stores calendar attributes such as Date, Month, Month Number, and Quarter**.**

6.4.3 Cardinality & Relationships:

Cardinality	Relationship
Many → One	Fact_Mandi_Erode[Commodity_Code] → Dim_Commodity[Commodity_Code]
Many → One	Fact_Mandi_Erode[Arrival_Date] → DateTable[Date]
Many → One	Fact_Mandi_Erode[Market] → Dim_Market[Market]

This modelling approach improves reporting performance, supports slicing and drill-down, and enables efficient commodity, market, and time-based analysis.

7. DAX Measures, Formula & Purpose:

To support dashboard interactivity, KPI computation, time-based comparisons, and volatility analysis, several DAX measures were created in Power BI. These measures were used to summarize modal prices, calculate price spread, compare month-over-month and year-over-year movement, identify volatile commodities and markets, and detect better market opportunities.

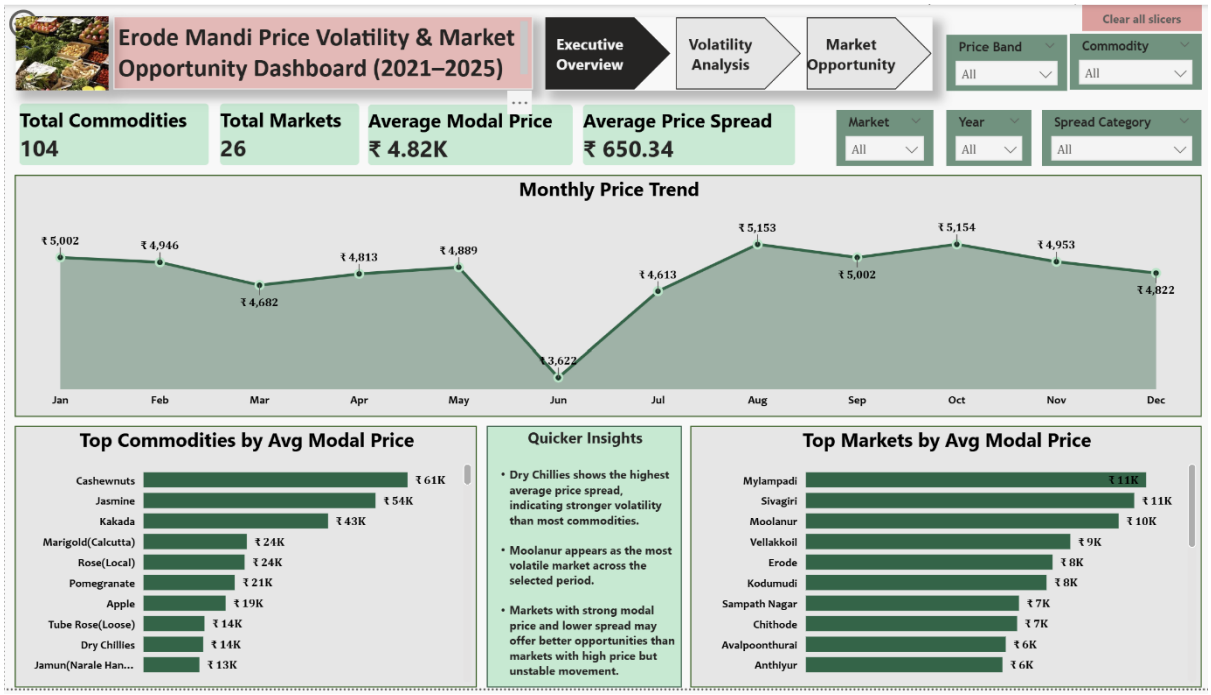
Measure Name	DAX Formula	Purpose
Total Commodities	`Total Commodities = DISTINCTCOUNT(Dim_Commodity[Commodity_Code])`	Calculates the total number of unique commodities available. Used in KPI cards.
Total Markets	`Total Markets = DISTINCTCOUNT(Dim_Market[Market])`	Calculates the total number of unique markets available. Used in KPI cards.
Average Modal Price	`Average Modal Price = AVERAGE(Fact_Mandi_Erode[Modal_Price])`	Calculates the overall average modal price. Used in KPI cards.
Average Min Price	`Average Min Price = AVERAGE(Fact_Mandi_Erode[Min_Price])`	Calculates the overall average of minimum price.
Average Max Price	`Average Max Price = AVERAGE(Fact_Mandi_Erode[Max_Price])`	Calculates the overall average of maximum price.
Average Price Spread	`Average Price Spread= [Average Max Price]-[Average Min Price]`	Calculates a summary spread indicator by measuring the difference between average maximum price and average minimum price.
Highest Commodity Spread	`Highest Commodity Spread = MAXX(ALLSELECTED(Dim_Commodity[Commodity]),[Average Price Spread])`	Identifies the highest average spread among commodities. Used in KPI cards.
Most Volatile Commodity	`Most Volatile Commodity = MAXX(TOPN(1, SUMMARIZE(Fact_Mandi_Erode, Fact_Mandi_Erode[Commodity], "AvgSpread", [Average Price Spread]), [AvgSpread], DESC), Fact_Mandi_Erode[Commodity])`	Returns the name of the commodity with the highest average price spread. Used in KPI cards.
Most Volatile Market	`Most Volatile Market = MAXX(TOPN(1, SUMMARIZE(Fact_Mandi_Erode, Fact_Mandi_Erode[Market], "AvgSpread", [Average Price Spread]), [AvgSpread], DESC), Fact_Mandi_Erode[Market])`	Returns the market with the highest average price spread. Used in KPI cards.

Previous Month Avg Modal Price	`Previous Month Avg Modal Price = CALCULATE([Average Modal Price], DATEADD(DateTable[Date], -1, MONTH))`	Calculates the previous month's average modal price for month-over-month comparison.
Previous Year Avg Modal Price	`Previous Year Avg Modal Price = CALCULATE([Average Modal Price], DATEADD(DateTable[Date], -1, YEAR))`	Calculates the same period's average modal price from the previous year for year-over-year comparison.
Month-Over-Month Change	`Month-Over-Month Change = [Average Modal Price] - [Previous Month Avg Modal Price]`	Measures change in modal price compared to the previous month. Used for opportunity analysis.
Year-Over-Year Change	`Year-Over-Year Change = [Average Modal Price] - [Previous Year Avg Modal Price]`	Measures change in modal price compared to the previous year. Used for opportunity analysis.
Highest MoM Gain	`Highest MoM Gain = MAXX(VALUES(Dim_Commodity[Commodity]), CALCULATE([Month-Over-Month Change]))`	Returns the highest month-over-month gain among commodities. Used in KPI cards.
Highest YoY Gain	`Highest YoY Gain = MAXX(VALUES(Dim_Commodity[Commodity]), CALCULATE([Year-Over-Year Change]))`	Returns the highest year-over-year gain among commodities. Used in KPI cards.
Top Opportunity Commodity	`Top Opportunity Commodity = MAXX(TOPN(1, SUMMARIZE(Dim_Commodity, Dim_Commodity[Commodity],"OppScore", [Market Opportunity Score]), [OppScore], DESC), Dim_Commodity[Commodity])`	Returns the commodity with the highest opportunity score.
Top Opportunity Market	`Top Opportunity Market = MAXX(TOPN(1, SUMMARIZE(Fact_Mandi_Erode, Dim_Market[Market], "OppScore", [Market Opportunity Score]), [OppScore], DESC), Dim_Market[Market])`	Returns the market with the highest opportunity score.
Market Opportunity Score	`Market Opportunity Score = DIVIDE([Average Modal Price], [Average Price Spread] + 1)`	This measure gives a higher score to markets where the average modal price is high and the price spread is low. This helps identify better opportunities with stronger prices and lower volatility.
Top 10 Volatile Markets	`Volatile Market Rank = RANKX(ALL(Dim_Market[Market]), [Average Price Spread], , DESC)`	Ranks markets by average spread. Used to show top volatile markets.
Top 10 Volatile Commodities	`Volatile Commodity Rank = RANKX(ALL(Dim_Commodity[Commodity]),[Average Price Spread],,DESC)`	Ranks commodities by average price spread. Used in Top 10 volatile commodities chart.

9. Analysis and Visualizations:

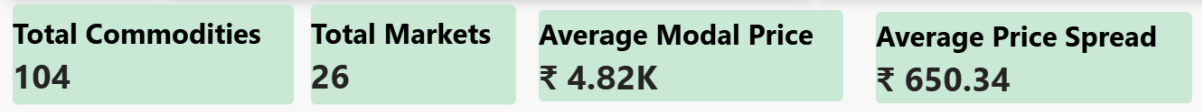
9.1 Executive Overview Dashboard:

Figure 2: Executive Dashboard



The Executive Overview page was designed to answer broad business questions related to overall market behaviour, price movement, premium commodities, and premium markets. The Price Band classification helps distinguish premium commodities and markets from moderate and lower price groups. It gives a consolidated view of the mandi dataset before deeper volatility and opportunity analysis.

Visual 1: KPI Cards:



Business Problem Addressed:

What is the overall scope of the mandi dataset, and what is the general price and spread behaviour?

Analysis:

The KPI cards summarize the number of commodities, number of markets, average modal price, and average price spread in the selected dataset.

Insight:

These indicators provide an immediate understanding of dataset coverage, prevailing market price level, and overall price variation.

Business Solution:

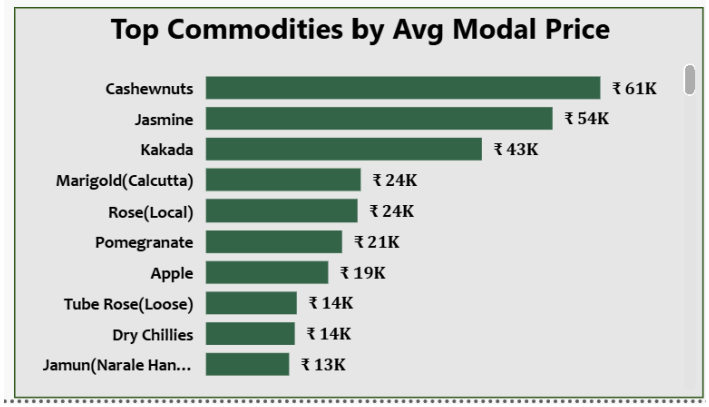
Users can use these summary indicators as a starting point to understand overall market conditions before moving into detailed volatility or opportunity analysis.

Visual 2 : Monthly Price Trend (Area Chart):

Business Problem Addressed:
Study seasonal and monthly price movement.
Analysis:
The area chart shows that average modal price remains relatively stable from January to May, declines sharply in June, and recovers strongly in August and October.
Insight:
- June appears to be the weakest pricing month.
- August and October show stronger price realization, indicating seasonal opportunity windows.

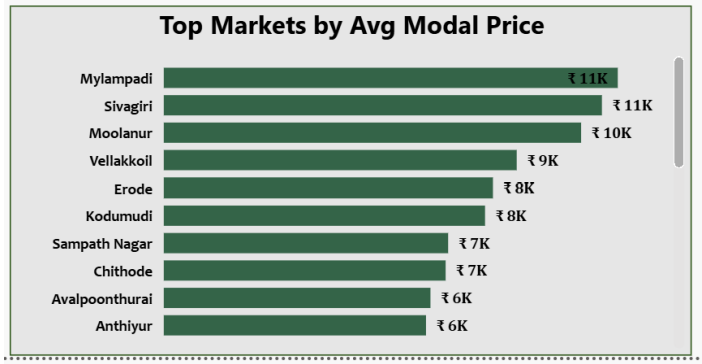
Business Solution:
Farmers and traders can improve price realization by monitoring seasonal trends and avoiding weak-selling months where possible.

Visual 3 : Top Commodities by Avg Modal Price (Bar Chart):



Business Problem Addressed:
Identify high-price and low-price commodities.
Analysis:
The chart shows that Cashewnuts, Jasmine, and Kakada have the highest average modal prices among the commodities in the dataset.
Insight:
A small group of commodities dominates the premium pricing segment, indicating stronger value potential than the rest.
Business Solution:
Producers and traders can prioritize premium commodities for higher-value selling strategies while also considering spread and volatility before making decisions.

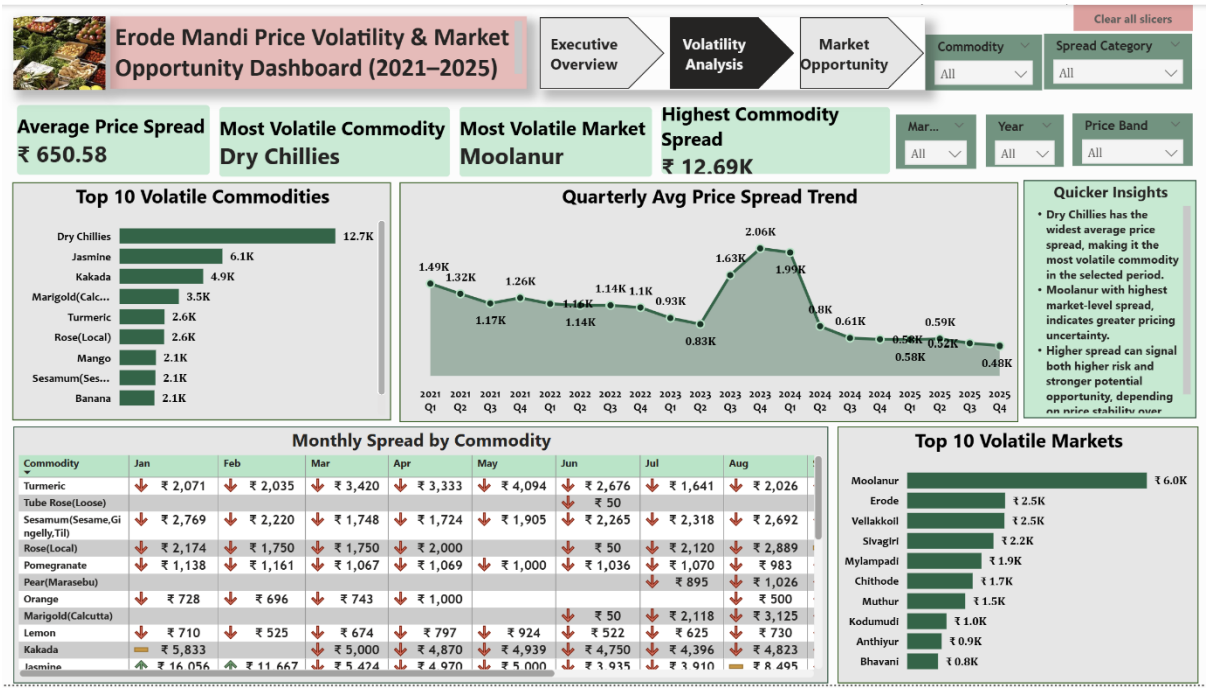
Visual 4 : Top Markets by Avg Modal Price (Bar Chart):



Business Problem Addressed:
Identify high-price and low-price markets and compare modal price across markets.
Analysis:
The market comparison shows that Mylampadi, Sivagiri, and Moolanur provide the highest average modal prices among the markets in the dataset.
Insight:
Price realization differs significantly by market, which means location plays an important role in selling opportunity.
Business Solution:
Sellers should compare market-level prices before choosing where to sell and prioritize markets that consistently provide stronger average modal prices.

9.2 Volatility Analysis Dashboard:

Figure 3: Volatility Analysis Dashboard



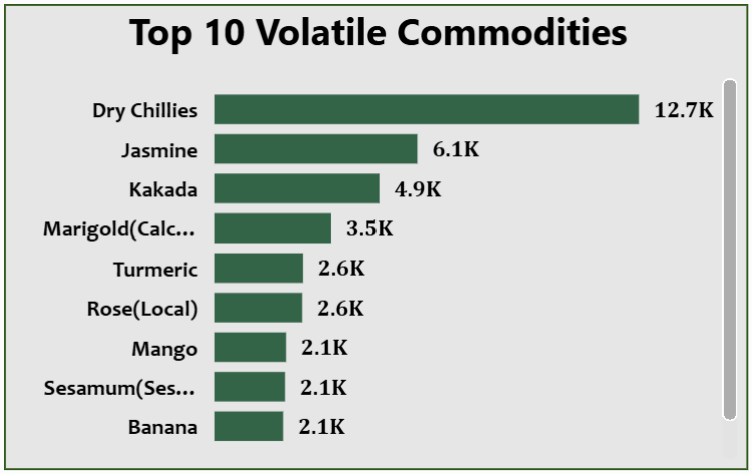
The Volatility Analysis page was created to answer business questions related to commodity instability, unstable markets, price spread behaviour, and the identification of stable versus unstable market conditions. The calculated fields Price Spread and Spread Category support the interpretation of stable and unstable commodities and markets.

Visual 1: Volatility KPI Cards

Business Problem Addressed:
Which commodities and markets are the most unstable, and how strong is the spread in the data?
Analysis:
The KPI cards summarize average spread, the most volatile commodity, the most volatile market, and the highest observed commodity spread.
Insight:
These indicators quickly identify where market instability and pricing uncertainty are concentrated.

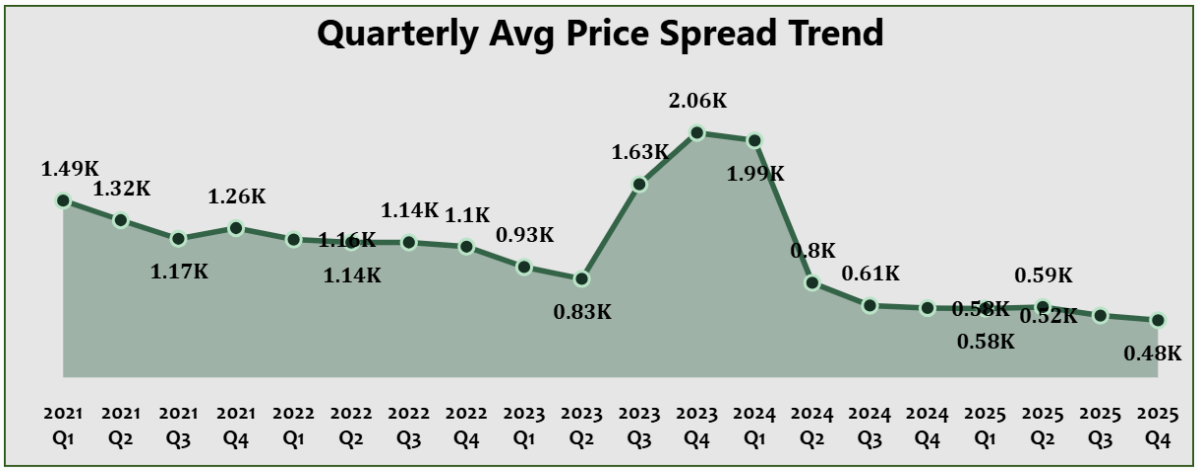
Business Solution:
Traders and analysts can use these summary metrics to focus attention on high-risk commodities and markets before making selling decisions.

Visual 2: Top 10 Volatile Commodities:



Business Problem Addressed:
Analyze commodity-wise price volatility.
Analysis:
The chart ranks commodities by average price spread, highlighting those with the widest difference between minimum and maximum prices.
Insight:
Commodities such as Dry Chillies show significantly wider spread, indicating higher volatility and greater pricing uncertainty.
Business Solution:
Volatile commodities should be monitored carefully, and sellers should consider both potential reward and risk before trading them.

Visual 3: Quarterly Average Price Spread Trend:



Business Problem Addressed:
Study volatility over time.
Analysis:
The quarterly trend shows how price spread changes across periods, helping identify phases of higher and lower market instability.

Insight:
Certain periods show consistently higher spread, suggesting that volatility may have a seasonal or recurring component.
Business Solution:
Monitoring spread over time can help traders avoid unstable periods and identify more favourable timing for selling.

Visual 4: Monthly Spread by Commodity Matrix:

Back to report | MONTHLY SPREAD BY COMMODITY

Commodity	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Turmeric	↓ ₹ 2,071	↓ ₹ 2,035	↓ ₹ 3,420	↓ ₹ 3,333	↓ ₹ 4,094	↓ ₹ 2,676	↓ ₹ 1,641	↓ ₹ 2,026	↓ ₹ 1,971	↓ ₹ 2,578	↓ ₹ 2,595	↓ ₹ 2,815
Tube Rose(Loose)						↓ ₹ 50				↓ ₹ 3,500		
Sesamum(Sesame,Gingelly,Til)	↓ ₹ 2,769	↓ ₹ 2,220	↓ ₹ 1,748	↓ ₹ 1,724	↓ ₹ 1,905	↓ ₹ 2,265	↓ ₹ 2,318	↓ ₹ 2,692	↓ ₹ 2,537	↓ ₹ 1,653	↓ ₹ 1,929	↓ ₹ 2,446
Rose(Local)	↓ ₹ 2,174	↓ ₹ 1,750	↓ ₹ 1,750	↓ ₹ 2,000		↓ ₹ 50	↓ ₹ 2,120	↓ ₹ 2,889	— ₹ 6,333	↓ ₹ 4,243	↓ ₹ 3,667	↓ ₹ 2,000
Pomegranate	↓ ₹ 1,138	↓ ₹ 1,161	↓ ₹ 1,067	↓ ₹ 1,069	↓ ₹ 1,000	↓ ₹ 1,036	↓ ₹ 1,070	↓ ₹ 983	↓ ₹ 992	↓ ₹ 1,009	↓ ₹ 970	↓ ₹ 1,027
Pear(Marasebu)							↓ ₹ 895	↓ ₹ 1,026	↓ ₹ 929	↓ ₹ 944		
Orange	↓ ₹ 728	↓ ₹ 696	↓ ₹ 743	↓ ₹ 1,000				↓ ₹ 500	↓ ₹ 550	↓ ₹ 655	↓ ₹ 856	↓ ₹ 964
Marigold(Calcutta)						↓ ₹ 50	↓ ₹ 2,118	↓ ₹ 3,125	↓ ₹ 5,455	↓ ₹ 4,118	↓ ₹ 3,385	↓ ₹ 1,667
Lemon	↓ ₹ 710	↓ ₹ 525	↓ ₹ 674	↓ ₹ 797	↓ ₹ 924	↓ ₹ 522	↓ ₹ 625	↓ ₹ 730	↓ ₹ 856	↓ ₹ 801	↓ ₹ 701	↓ ₹ 673
Kakada	— ₹ 5,833		↓ ₹ 5,000	↓ ₹ 4,870	↓ ₹ 4,939	↓ ₹ 4,750	↓ ₹ 4,396	↓ ₹ 4,823	↓ ₹ 4,815	↓ ₹ 4,844	— ₹ 5,694	↓ ₹ 5,063
Jasmine	↑ ₹ 16,056	↑ ₹ 11,667	↓ ₹ 5,424	↓ ₹ 4,970	↓ ₹ 5,000	↓ ₹ 3,935	↓ ₹ 3,910	— ₹ 8,495	↓ ₹ 4,864	↓ ₹ 5,188	— ₹ 5,941	— ₹ 6,400
Jamun(Narale Hannu)				↓ ₹ 500		↓ ₹ 1,056	↓ ₹ 1,000	↓ ₹ 1,000				
Ground Nut Seed			↓ ₹ 720	↓ ₹ 400				↓ ₹ 250	↓ ₹ 500			
Green Peas	↓ ₹ 768	↓ ₹ 764	↓ ₹ 600	↓ ₹ 800	↓ ₹ 1,000		↓ ₹ 1,000	↓ ₹ 1,020	↓ ₹ 1,000	↓ ₹ 900	↓ ₹ 1,000	↓ ₹ 991
Green Gram Dal (Moong Dal)	↓ ₹ 2,924	↓ ₹ 453	↓ ₹ 1,990								↓ ₹ 1	
Grapes	↓ ₹ 2,226	↓ ₹ 1,179	↓ ₹ 1,000	↓ ₹ 1,397	↓ ₹ 690	↓ ₹ 679	↓ ₹ 792	↓ ₹ 1,000	↓ ₹ 864	↓ ₹ 633	↓ ₹ 796	↓ ₹ 700
Ginger(Green)	↓ ₹ 559	↓ ₹ 491	↓ ₹ 473	↓ ₹ 556	↓ ₹ 560	↓ ₹ 385	↓ ₹ 617	↓ ₹ 651	↓ ₹ 635	↓ ₹ 677	↓ ₹ 539	↓ ₹ 487
Garlic		↓ ₹ 1,125							↓ ₹ 400			
Dry Chillies	— ₹ 11,174	↑ ₹ 12,242	— ₹ 11,410	— ₹ 10,524	— ₹ 10,733	— ₹ 11,531	↑ ₹ 17,225	↑ ₹ 13,730	↑ ₹ 14,215	↑ ₹ 15,479	↑ ₹ 14,635	↑ ₹ 12,848
Cowpea (Lobia/Karamani)	↓ ₹ 1,631	↓ ₹ 1,968	↓ ₹ 1,026	↓ ₹ 2,696				↓ ₹ 200	↓ ₹ 1,300	↓ ₹ 850	↓ ₹ 1,573	↓ ₹ 2,396
Cotton	↓ ₹ 1,696	↓ ₹ 1,121	↓ ₹ 1,358	↓ ₹ 1,792	↓ ₹ 2,076	↓ ₹ 2,679	↓ ₹ 2,044	↓ ₹ 1,632	↓ ₹ 1,709	↓ ₹ 1,467	↓ ₹ 1,388	↓ ₹ 1,528
Copra	↓ ₹ 1,771	↓ ₹ 1,759	↓ ₹ 1,719	↓ ₹ 1,599	↓ ₹ 1,665	↓ ₹ 1,572	↓ ₹ 1,519	↓ ₹ 1,298	↓ ₹ 1,401	↓ ₹ 1,419	↓ ₹ 1,581	↓ ₹ 1,691
Cashewnuts	↓ ₹ 904											
Black Gram (Urd Beans)(Whole)	↓ ₹ 540	↓ ₹ 924	↓ ₹ 733								↓ ₹ 1	↓ ₹ 250
Apple	↓ ₹ 1,000	↓ ₹ 1,000	↓ ₹ 1,000	↓ ₹ 1,000	↓ ₹ 1,000	↓ ₹ 1,000	↓ ₹ 1,702	↓ ₹ 978	↓ ₹ 932	↓ ₹ 955	↓ ₹ 1,000	↓ ₹ 1,000

Business Problem Addressed:
Which commodities repeatedly show higher spread across months?

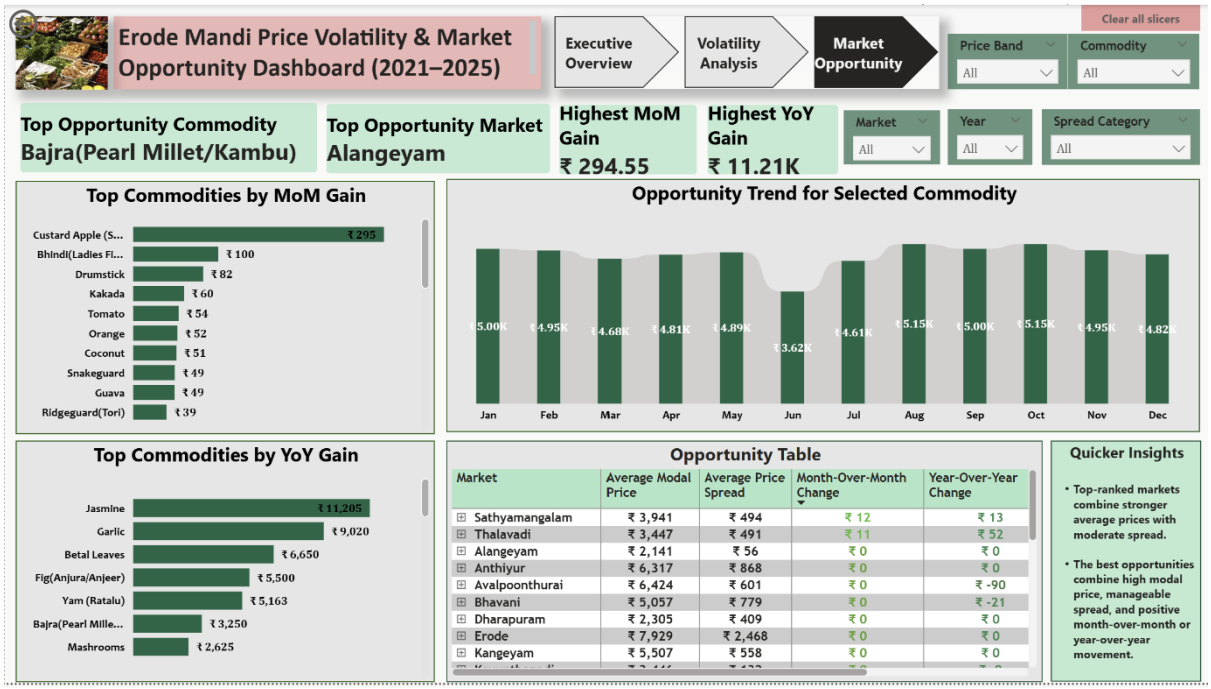
Analysis:
The matrix compares spread values across commodities and months to identify recurring patterns of instability.
Insight:
It reveals which commodities show repeated high spread and which remain relatively stable across time.
Business Solution:
Users can use this view to identify commodities that require more careful timing and risk-based selling decisions.

Visual 5: Top 10 Volatile Markets:

Business Problem Addressed:
Identify stable and unstable markets.
Analysis:
The chart ranks markets by average price spread, helping identify where market-level instability is concentrated.
Insight:
Markets with wider spread represent greater uncertainty and lower pricing stability.
Business Solution:
Sellers should balance high price with stability and avoid relying solely on premium markets if spread is too high.

9.3 Market Opportunity Dashboard:

Figure 4: Market Opportunity Dashboard



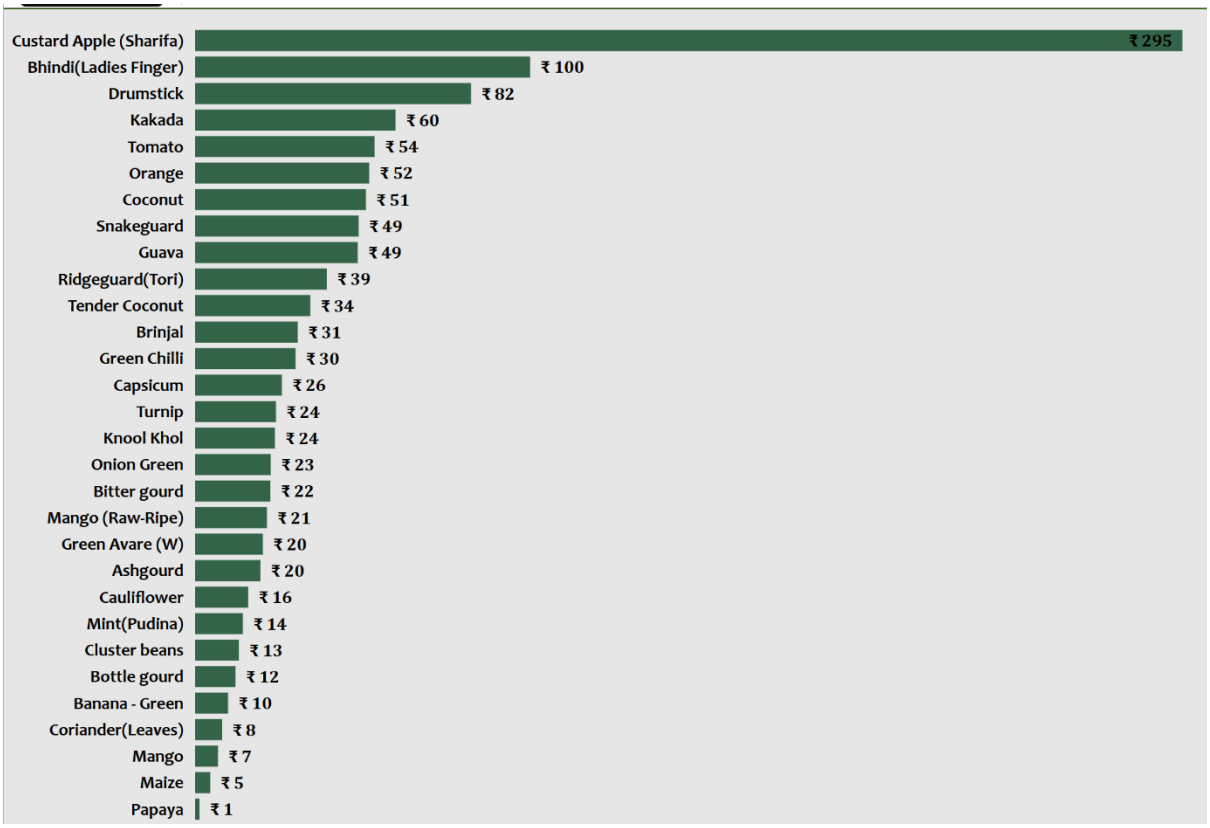
The Market Opportunity page was designed to answer the final business question: which commodities and markets represent better selling opportunities when price level, recent movement, and spread are considered together

Visual 1: Opportunity KPI Cards:

Business Problem Addressed:
Which commodities and markets currently show the strongest opportunity signals?
Analysis:
The KPI cards highlight the top opportunity commodity, top opportunity market, highest month-over-month gain, and highest year-over-year gain.
Insight:
These indicators quickly identify where favourable short-term and medium-term market conditions are strongest.

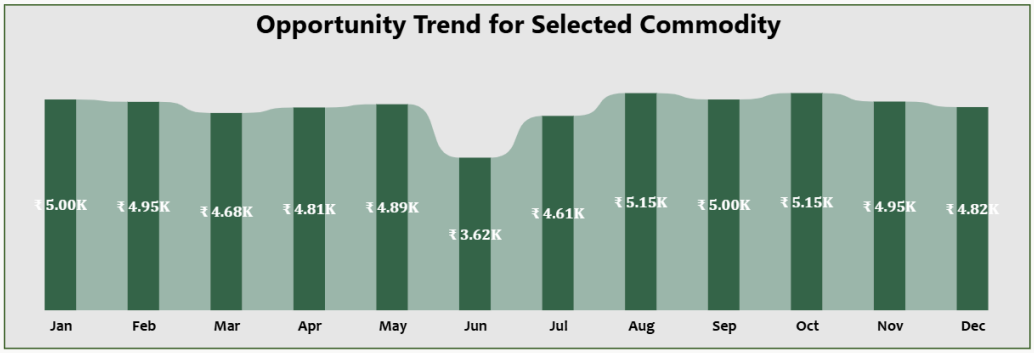
Business Solution:
Users can use these KPIs to prioritize commodities and markets that combine stronger prices with positive momentum.

Visual 2: Top Commodities by MoM Gain:



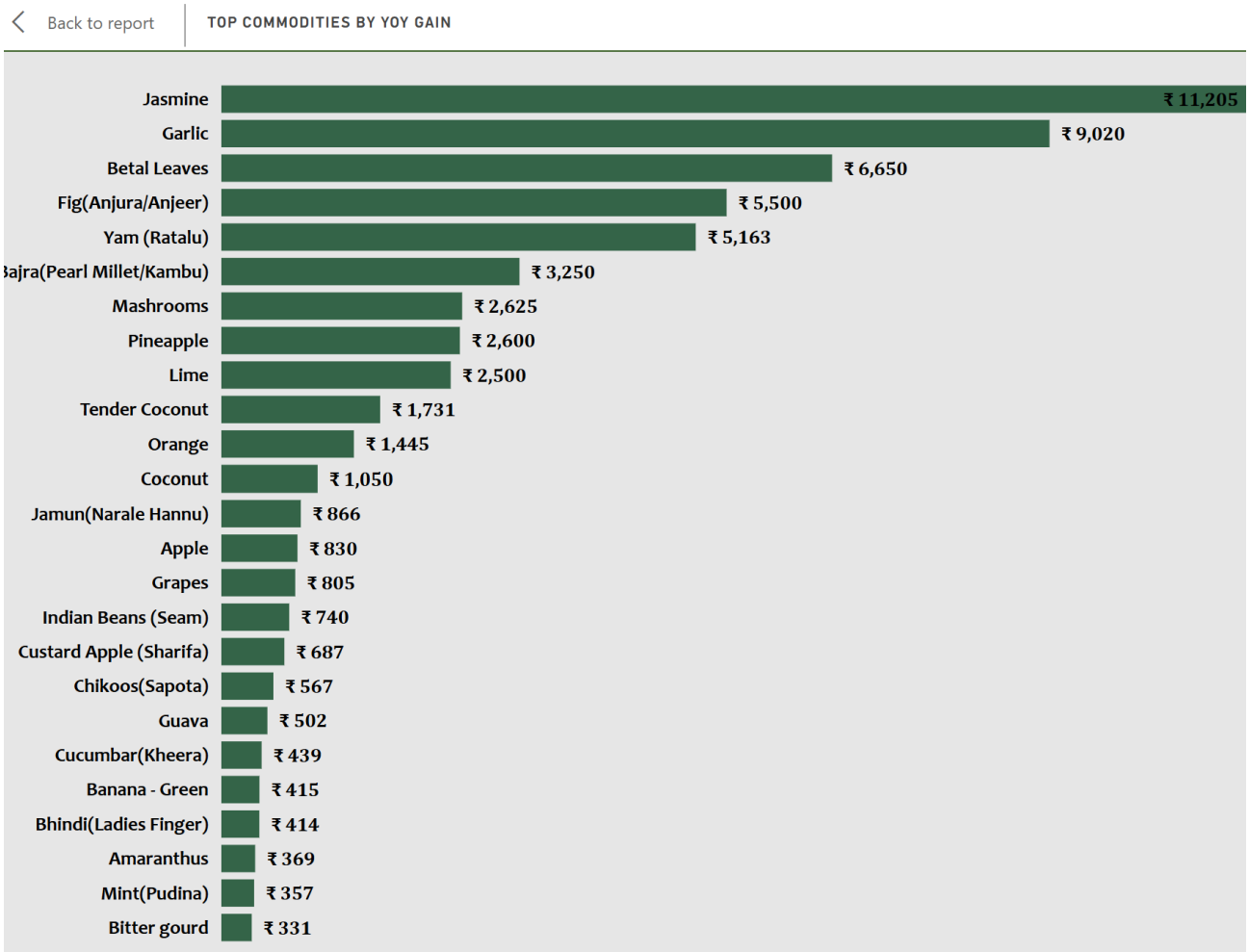
Business Problem Addressed:
Identify short-term market opportunity.
Analysis:
The chart compares month-over-month price movement across commodities to identify recent gainers.
Insight:
Commodities with stronger positive monthly movement may represent immediate opportunity in the short term.
Business Solution:
Traders can monitor short-term gainers to identify commodities with recent positive momentum.

Visual 3: Opportunity Trend for Selected Commodity:



Business Problem Addressed:
Compare current movement with previous month and previous year.
Analysis:
The multi-line trend compares current average modal price against previous month and previous year values for the selected commodity.
Insight:
This helps determine whether the selected commodity is improving, weakening, or stabilizing over time.
Business Solution:
Users can use this comparison to judge whether the current market condition supports immediate selling or continued monitoring.

Visual 4: Top Commodities by YoY Gain:



Business Problem Addressed:
Which commodities have shown strong year-over-year price improvement?
Analysis:
The chart compares year-over-year gain across commodities to identify those with stronger long-term price performance.
Insight:
It highlights commodities with better long-term opportunity and stronger annual movement.
Business Solution:
Traders can use this comparison to identify commodities with stronger medium-term and long-term potential.

Visual 5: Opportunity Table:

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Market	Average Modal Price	Average Price Spread	Month-Over-Month Change	Year-Over-Year Change	Volatile Commodity Rank
Sathyamangalam	₹ 3,941	₹ 494	₹ 12	₹ 13	21
Thalavadi	₹ 3,447	₹ 491	₹ 11	₹ 52	11
Alangeyam	₹ 2,141	₹ 56	₹ 0	₹ 0	5
Anthiyur	₹ 6,317	₹ 868	₹ 0	₹ 0	9
Avalpoonthurai	₹ 6,424	₹ 601	₹ 0	₹ -90	3
Bhavani	₹ 5,057	₹ 779	₹ 0	₹ -21	5
Dharapuram	₹ 2,305	₹ 409	₹ 0	₹ 0	11
Erode	₹ 7,929	₹ 2,468	₹ 0	₹ 0	2
Kangeyam	₹ 5,507	₹ 558	₹ 0	₹ 0	4
Kavunthapadi	₹ 3,446	₹ 132	₹ 0	₹ -9	4
Kodumudi	₹ 7,739	₹ 991	₹ 0	₹ -9	2
Kunnathur	₹ 2,525	₹ 415	₹ 0	₹ 0	1
Moolanur	₹ 10,054	₹ 6,029	₹ 0	₹ 0	2
Muthur	₹ 6,176	₹ 1,466	₹ 0	₹ 0	3
Mylampadi	₹ 10,936	₹ 1,837	₹ 0	₹ -52	2
Nambiyur	₹ 3,164	₹ 163	₹ 0	₹ 0	2
Punchaipuliyampatti	₹ 5,132	₹ 516	₹ 0	₹ 0	4
Sivagiri	₹ 10,555	₹ 2,178	₹ 0	₹ 0	2
Vellakkoil	₹ 8,502	₹ 2,450	₹ 0	₹ 0	2
Boothapadi	₹ 4,294	₹ 541	₹ -9	₹ -190	8
Periyar Nagar	₹ 4,047	₹ 509	₹ -11	₹ 4	19
Perundurai	₹ 4,494	₹ 574	₹ -15	₹ -384	6
Chithode	₹ 6,790	₹ 1,654	₹ -16	₹ -121	2
Gobichettipalayam	₹ 4,785	₹ 566	₹ -21	₹ -774	14
Elumathur	₹ 5,728	₹ 465	₹ -30	₹ -30	2
Sampath Nagar	₹ 6,850	₹ 762	₹ -77	₹ -355	21

Business Problem Addressed:
Identify favourable selling markets.
Analysis:
The table combines average modal price, spread, month-over-month change, and year-over-year change for detailed comparison across markets.
Insight:
The best opportunities are found where stronger price realization is supported by manageable spread and positive movement.
Business Solution:
Sellers can use this table as a decision-support view to identify better markets instead of relying only on a single price value.

Thus, each dashboard page was aligned to a specific part of the business problem. The Executive Overview answers questions related to overall pricing and premium commodities and markets. The Volatility Analysis page addresses stability, spread, and uncertainty across commodities and markets. The Market Opportunity page combines price level, movement, and spread to support better selling decisions.

10. Insights & Conclusions:

The analysis of Erode mandi price data from 2021 to 2025 helped identify important patterns related to commodity pricing, market-level variability, seasonal movement, and market opportunity. The dashboard supports agricultural market intelligence by transforming raw mandi data into actionable insights for comparison and decision-making.

10.1 Key Findings:
The major findings from the Erode mandi analysis are summarized below:

- The dataset covers **104 commodities** across **26 markets**, indicating broad mandi coverage within Erode district.
- The overall **average modal price is approximately ₹4.82K**, while the **average price spread is around ₹650**, showing moderate variation in market prices.
- **Cashewnuts, Jasmine, and Kakada** emerge as premium commodities with consistently higher average modal prices.
- **Mylampadi, Sivagiri, and Moolanur** are identified as the leading premium markets in terms of average price realization.
- **Dry Chillies** records the highest average price spread, making it the most volatile commodity in the analysis.
- **Moolanur** appears as the most volatile market, indicating greater price uncertainty compared to other markets.
- Monthly price trends suggest that price realization is not uniform throughout the year, pointing to possible seasonal selling opportunities.
- Better market opportunities are observed where strong modal price is supported by manageable spread and positive recent movement.

10.2 Descriptive Insights:

Descriptive analysis shows that mandi prices vary notably across commodities, markets, and time periods. A small number of commodities remain in premium price bands, while others operate within moderate or lower price ranges. Market-wise comparison confirms that price realization differs significantly by location, and monthly trend analysis suggests that timing influences selling outcomes. Spread-based analysis also indicates that high price does not always imply stability.

10.3 Diagnostic Insights:

Diagnostic analysis suggests that wider price spread reflects greater variation between minimum and maximum price, indicating higher volatility and lower pricing stability. Commodities such as Dry Chillies and markets such as Moolanur exhibit wider spread, which may be associated with supply-demand imbalance, seasonal factors, or local market variability. Premium markets with high spread may still offer opportunity, but they also involve greater uncertainty.

10.4 Predictive Insights:

Based on historical patterns, commodities and markets that repeatedly exhibit high spread may continue to show greater volatility unless conditions stabilize. Seasonal movement suggests that certain months may remain more favourable for selling, while commodities with positive month-over-month and year-over-year change may continue to offer short-term opportunity.

10.5 Prescriptive Insights:

The findings suggest that market decisions should not be based on high price alone. Instead, users should combine modal price, spread, and recent movement to identify better selling opportunities. Stable high-price markets should be preferred over highly volatile premium markets, and commodities with recurring seasonal peaks should be monitored more closely for timing-based selling decisions.

11. Conclusion:

The integration of Excel and Power BI proved effective for end-to-end agricultural market analysis, from raw mandi data cleaning and transformation to interactive dashboard reporting. This mini project successfully identified commodity-wise price volatility, premium and low-price markets, seasonal price movement, and market opportunity patterns using Erode mandi data from 2021 to 2025.

The dashboard was developed through a structured analytical approach consisting of three perspectives: **Executive Overview, Volatility Analysis, and Market Opportunity**. Together, these views addressed the core business problem by helping users understand what to sell, where to sell, and when better market conditions may exist.

Overall, the project demonstrates how real-world government data can be transformed into a decision-oriented agricultural market intelligence solution that supports clearer comparison, trend analysis, and informed market decision-making.

12. References:

1) Open Government Data Platform India / AGMARKNET
2) Microsoft Excel and Power Query
3) Microsoft Power BI Documentation

13. Limitations:

Although the project successfully delivers meaningful agricultural market intelligence using Erode mandi data, a few limitations must be acknowledged.

- The analysis is limited to the mandi price records available for **Erode district** and may not represent pricing behaviour in other districts or states.
- The project primarily uses **price-based attributes** such as minimum price, maximum price, and modal price. Important factors such as **commodity arrivals, quantity traded, transportation cost, rainfall, and storage conditions** were not included in the model.
- The opportunity analysis is based on **historical pricing patterns and spread behaviour** and should not be interpreted as a guaranteed future outcome.
- Price spread has been used as a proxy for market volatility; however, it does not fully capture all real-world factors influencing market risk.
- The dashboard focuses on descriptive and diagnostic insights and does not include advanced forecasting or predictive machine learning models.

14. Future Scope:

This project can be extended in several ways to make the analysis more comprehensive and decision-oriented.

- Include **commodity arrivals and traded quantity data** to improve opportunity analysis and demand-supply interpretation.
- Expand the study beyond Erode district to include **multi-district or state-level mandi comparison**.
- Add **forecasting techniques** to predict future price movement for selected commodities.
- Incorporate **weather, rainfall, festival season, and transportation factors** to better explain seasonal price behaviour.
- Develop a **farmer-oriented recommendation page** that suggests better selling months and markets for specific commodities.
- Enhance the dashboard with **drill-through pages, tooltip pages, and dynamic recommendation logic** for more detailed decision support.
- Integrate the model with **real-time or frequently refreshed government data** to make the dashboard more actionable for live monitoring.